Saturated Superfluid Vacuum IX
CMB and the Primordial Phonon Bath

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Draft

Abstract

This paper offers a single, deliberately narrow reading of the cosmic microwave background (CMB) within the Saturated Superfluid Vacuum (SSV) framework: that the CMB is acoustic noise integrated from the reconnection events of the void-to-saturation transition (Stages 4–5 of Paper VIII [1]), rather than a separately postulated radiation-era relic. The medium that survives that transition is not quiet; it carries a residual bath of phonons — the Goldstone sound modes of the saturated condensate [2] — whose present-day, red-shifted spectrum is what an observer reads as the CMB.

The paper is explicitly *interpretive* in its central claim and honest about what it does *not* establish. It does *not* derive the CMB temperature $T_0 = 2.725$ K, does *not* derive the blackbody form of the spectrum, and does *not* compute the spectral distortions; each of those is marked with an explicit gap box and tied to the same missing input — the absolute saturation pressure $P_0(\rho_0)$ — that blocks the cosmological-constant value in Paper VIII. What the paper inherits is Paper VIII's exploratory Kibble–Zurek defect scan, whose fitted exponent does not yet confirm the expected scaling. Every claim is tagged *derived*, *structural*, *candidate*, *interpretive*, or *deferred* in the Discussion (§7), with explicit criteria for what observational or theoretical work would change its status. The purpose is to state the phonon-bath reading of the CMB cleanly and to fix the support conditions for it — not to present a quantitative CMB cosmology.

1 Introduction

The cosmic microwave background is, in the standard account, a relic of a hot radiation era: a thermal photon gas that decoupled from baryonic matter at recombination and has since free-streamed and cooled with the expansion. The SSV framework does not have a separate photon gas to begin with — photons are transverse Goldstone/Bogoliubov modes of the condensate [2], and the "radiation era" is not an independent postulate but the turbulent aftermath of the saturation transition described in Paper VIII [1]. This paper asks the narrow question that follows: *if the saturating vacuum necessarily produces a bath of sound modes, what does that bath look like today, and is it consistent with being read as the CMB?*

The scope is deliberately small. This paper consumes Paper VIII's cosmogonic sequence as input and does not reopen it; it borrows the entropy / arrow-of-time machinery of Paper III [3] for the irreversibility of the integrated noise; and it uses the equation-of-state language of Paper II [4] for the medium's acoustic response. It does *not* address inflationary phenomenology, structure formation (galactic scales live in Paper VI), late-time acceleration (deferred to Paper VII-b and to Paper VIII's prediction C2), or any particle and force-sector content (Papers I–IV). The material developed here is the CMB and primordial-phonon content that issue #10 moved out of Paper III; this draft is its destination, written under issue #100.

Every claim below is tagged in the Discussion (§7). The reader should treat the central reading — CMB as integrated reconnection noise — as *interpretive*, and should read the temperature, blackbody character, and distortion content as *deferred* (gap-boxed), not as results.

2 The Primordial Phonon Bath

Paper VIII, Stage 4, ends with the saturation fronts colliding and the binding energy released into the kinetic energy of the flow field, "driving turbulence." Stage 5 then hypothesises defect nucleation during phase ordering. If the postulated saturation quench occurs, it can produce both defects and sound; neither their cosmological abundance nor their identification with matter and radiation is derived. This paper explores the second product.

In a saturated superfluid the low-energy excitations above the ground state ρ_0 are phonons: linear sound modes with dispersion $\omega = c_s k$ in the long-wavelength limit, where c_s is the medium's sound speed fixed by the equation of state $P(\rho)$ of Paper II [4]. The saturation transition is postulated to be non-adiabatic and exothermic, so it could leave the condensate in an excited state populated by these modes. We call this candidate population the *primordial phonon bath*.

> *Claim (§2).* The saturated vacuum is left, after the Stage 4–5 transition, with a residual bath of phonon (sound) modes carrying the energy shed by the relaxing flow field. Status: *interpretive*; neither the quench dynamics, bath abundance nor spectrum is derived here.

A joint defect-and-phonon simulation could test whether frozen defects and sound carry a reproducible ratio. Paper VIII currently counts only defects; it does not compute the baryon-to-photon ratio η.

The bath is not load-bearing for gravity (#119 outcome). During the #119 gravity-sector redesign, this bath was the candidate common drive for a bath-driven (secondary-Bjerknes) gravitational mechanism — which, had it survived, would have made the present paper's object load-bearing for the gravity sector, with G inheriting the bath amplitude. The pre-registered test returned a negative: the bath-driven force between passive defects is sign-definite and frequency-blind, as required, but short-ranged — the wrong range for Newtonian gravity (#119 Phase 2). In the revised architecture the gravitational source is each defect's *intrinsic* two-term structure (#119 H7a/H8; Paper IV), and the phonon bath carries no gravitational load. The bath's status in this paper is therefore unchanged by the gravity-sector redesign, and nothing here depends on it.

3 The CMB as Integrated Reconnection Noise

The central reading of this paper is the following identification.

> *Central claim (§3, interpretive).* The cosmic microwave background is the present-day, red-shifted image of the primordial phonon bath: acoustic noise integrated over all the reconnection and relaxation events of the void-to-saturation transition, propagated forward through the expanding saturated medium.

Three features of the picture are worth stating precisely, because they are what a supporting computation would have to reproduce.

Integration, not a single surface. In the standard account the CMB comes from a single last-scattering surface. Here the noise is *integrated*: each reconnection event during Stage 5 radiates a sound pulse, and the bath is the superposition of an enormous number of such pulses with random phases. This is the sense in which the spectrum is expected to be smooth — it is the central-limit sum of many independent emitters — but smoothness of a noise spectrum is not the same as a Planck (blackbody) spectrum, a distinction we are careful about in §4.

Irreversibility. The integrated noise is irreversible: it is the wake entropy of Paper III [3] read at cosmological scale. The arrow of time that Paper III attributes to the growth of wake complexity is, here, the statement that the phonon bath does not spontaneously refocus into coherent flow. It may be a large wake-entropy reservoir, but this paper contains no entropy inventory comparing it with black holes or other cosmic components.

Redshift as medium relaxation. The cooling of the bath is the slow relaxation of the background ρ_0 and the corresponding stretching of mode wavelengths, the same background-relaxation channel that Paper VIII's prediction C2 ties to late-time acceleration. We do *not* import a metric expansion here; we note only that the qualitative redshift of the bath and the qualitative late-time relaxation are the same effect viewed at different epochs, and we defer the quantitative expansion history to Paper VII-b.

This section is the reason the paper exists, and it is entirely *interpretive*. It reads a known observable (a near-isotropic microwave bath) as a known SSV object (the Goldstone sound field of a quenched condensate). It does not, by itself, predict any number.

4 Spectrum, Temperature, and the Equation of State

The interpretive reading of §3 becomes physics only if the bath can be shown to carry the right spectrum and temperature. This is where the paper is most honest about its limits.

4.1 Why a thermal spectrum is plausible but not derived

A blackbody (Planck) spectrum is the equilibrium distribution of a bosonic field that has had time to thermalise through mode–mode interactions. Phonons in a superfluid do interact (the GPE nonlinearity couples modes), so a saturated medium left in an excited phonon state will tend toward a thermal distribution. That is the plausibility argument. But "tends toward thermal" is not "is Planckian to one part in 10^5," which is what the measured CMB demands. Establishing the blackbody form requires a thermalisation (ergodicity) argument for the phonon bath that this paper does not provide.

Gap: blackbody character of the bath is not derived

The Planck form of the CMB spectrum is, here, *deferred*. It would follow from a demonstration that the primordial phonon bath thermalises — i.e. that the GPE phonon–phonon interactions drive the mode-occupation numbers to a Bose–Einstein distribution on a timescale short compared with the saturation-epoch expansion time. No such thermalisation calculation has been performed. Until it is, the "integrated reconnection noise" is established only as a *smooth* noise spectrum (central-limit superposition of many emitters), not as a *Planck* spectrum. Support criterion: a phonon-Boltzmann or wave-turbulence calculation in the saturated GPE showing relaxation to Bose–Einstein occupation.

4.2 Why the temperature is not derived

Even granting a thermal form, the absolute temperature $T_0 = 2.725$ K is set by the energy density deposited in the bath at saturation and by the subsequent dilution. The energy density at saturation is fixed by the depth of the equation of state — specifically by the absolute saturation pressure $P_0(\rho_0)$ — which is the *same* quantity that Paper VIII flags as the missing input blocking a numerical value of Λ, and that Papers II/IV/VII-b track as an open closure problem.

Gap: CMB temperature $T_0 = 2.725$ K is not derived

The present CMB temperature is *deferred*, and it is blocked by exactly the input that blocks Λ in Paper VIII [1] and the absolute energy scale in Papers II/IV: the absolute saturation pressure $P_0(\rho_0)$. Schematically the bath energy density at saturation is $u_{\rm sat} \sim P_0$, and T_0 follows from $u_{\rm sat}$ diluted by the expansion factor between the saturation epoch and now; neither P_0 nor the SSV expansion factor is derived in the series yet. The galactic sector now joins this list: issue #129 reframes Paper VI's underived gravitational coupling as an emergent acceleration scale $a_0 \sim c^2\sqrt{\Lambda} \sim cH_0$, so the *same* $P_0 \to \Lambda$ closure would simultaneously fix the galactic rotation-curve scale — closing both sectors at once, or failing both together. We therefore make *no* prediction for T_0. Support criterion: a derivation of $P_0(\rho_0)$ from the saturation equation of state (the Paper II/IV/VII-b closure item), *plus* the Paper VII-b emergent-expansion factor, which together would turn T_0 into a falsifiable SSV number.

4.3 What the equation of state does fix (structurally)

What the medium's equation of state *does* fix, without the absolute scale, is the *shape* relations: the sound speed $c_s(\rho)$, the dispersion's departure from linearity at the healing scale ξ (the Bogoliubov "roton-like" bend), and hence the wavelength at which the bath's spectrum must roll over relative to its peak. These are ratios, independent of P_0, and they are inherited directly from Paper II. They are *structural*: the framework fixes them in principle, but the present paper does not evaluate them numerically.

5 Statistical Departures from Blackbody

If the CMB is integrated reconnection noise rather than an equilibrium relic, the *departures* from a perfect blackbody are where the two pictures could be distinguished. In the SSV reading the distortions are not separately postulated mechanisms (energy injection, Silk damping, etc.) but signatures of the topological-defect statistics frozen in at saturation.

> *Claim (§5, candidate).* Spectral distortions of the CMB away from a pure Planck form encode the Kibble–Zurek defect statistics of Stage 5: the defect-density variance and correlation structure imprint a calculable, small, non-Planckian component on the integrated phonon spectrum.

This is a *candidate* claim with a definite falsification condition but no performed computation. Concretely, the spectrum should carry μ-type and y-type distortions whose amplitudes are tied to the same defect-density variance that sets η; the present observational bounds ($|\mu| \lesssim 9 \times 10^{-5}$, $|y| \lesssim 1.5 \times 10^{-5}$ from COBE/FIRAS [5]) are the falsification gate.

Gap: distortion amplitudes are not computed

The μ- and y-distortion amplitudes implied by the defect statistics are *deferred*. Computing them requires (i) the defect-density two-point statistics from a Kibble–Zurek simulation (the Paper VIII Stage 5 computation, already begun in `kibble_zurek_gpe`) carried to the level of correlation functions, and (ii) a map from those statistics to the phonon-bath occupation spectrum. Falsification condition: predicted $|\mu|, |y|$ exceeding the COBE/FIRAS bounds [5] would rule out the integrated-noise reading; predicted values far below current sensitivity would make the reading consistent but not yet testable. Support criterion: a defect-correlation $\to$ distortion-spectrum calculation with an explicit numerical μ, y.

6 Connection to SSV VIII Cosmogony

The exploratory quantitative input inherited from Paper VIII is a Kibble–Zurek-motivated defect scan. Stage 5 posits a defect density

$$n_{\text{defect}} \sim \frac{1}{\xi^3} \left(\frac{\tau_0}{\tau_Q} \right)^{d\nu/(1+\nu z)}, \tag{1}$$

with $\tau_0 = \xi/c$ the microscopic time and τ_Q the quench time of the saturation epoch. Paper VIII reports a stochastic time-dependent Ginzburg–Landau simulation (`kibble_zurek_gpe`) with a decreasing defect-count trend but a fitted exponent about 5.1 regression standard errors below the mean-field expectation. The scan therefore does not confirm the universality class. Values of τ_Q/τ_0 quoted there are obtained by inverting the observed $\eta = 6 \times 10^{-10}$; they are not independent predictions.

The simulation counts defects but does not count phonons, establish which defects become baryons, or derive their ratio. Writing $n_\gamma = n_{\text{baryon}}/\eta$ merely inserts the observed η and cannot serve as a prediction of the phonon-bath density.

> *Candidate link (§6).* A future joint calculation of defect and phonon production could predict their ratio. The present simulation supplies only exploratory defect counts, while the numerical η used here is observational input.

Gap: the entropy budget is not closed

The cosmological entropy budget — the photon (phonon-bath) entropy per baryon, of order η^{-1} only after importing the observed ratio — is *numerically* open, because the absolute photon entropy needs the bath temperature T_0, which is gap-boxed in §4. The arrow-of-time reading of Paper III [3] motivates treating the bath as a wake-entropy reservoir, but the present paper does not evaluate the budget in physical units. Support criterion: the same $P_0(\rho_0) + T_0$ closure required for the temperature.

7 Discussion: scope, status, and what would count as support

7.1 Claim taxonomy used in this paper

Following `numerical-minimisation-roadmap` in the repository, each claim is one of:

- *derived* — analytically reduced from the SSV Lagrangian and/or a converged numerical computation and *reused* here;
- *structural* — the dependence on the framework is clear and the result is structurally consistent, but the number depends on a computation deferred to another paper or workstream;
- *candidate* — a prediction stated here with a definite falsification condition but no supporting computation yet performed;
- *interpretive* — a synthesis or ontological statement that organises SSV results into a single picture without new quantitative content;
- *deferred* — a quantity that the picture requires but that is explicitly not evaluated here and is gap-boxed, with the blocking input named.

7.2 Claim status of this paper's content

- The primordial phonon bath (§2): *interpretive* — its *existence* follows structurally from a non-adiabatic saturation quench of a Goldstone sector; its spectrum is not derived.
- CMB as integrated reconnection noise (§3): *interpretive* — the central reading; predicts no number by itself.
- Blackbody character (§4): *deferred* (gap-boxed) — needs a phonon-thermalisation argument.
- CMB temperature T_0 (§4): *deferred* (gap-boxed) — blocked by the absolute saturation pressure $P_0(\rho_0)$, the same input that blocks Λ in Paper VIII.
- Equation-of-state shape relations (§4): *structural* — ratios inherited from Paper II, not evaluated here.
- Spectral distortions (§5): *candidate* — definite FIRAS falsification gate, amplitudes gap-boxed.
- Phonon-bath / defect link via η (§6): *candidate* — Paper VIII supplies an exploratory defect scan, not a phonon count or predicted ratio.
- Entropy budget (§6): *deferred* (gap-boxed) — structurally fixed by η, numerically blocked by T_0.

7.3 What observational or theoretical support would matter

1. *A phonon-thermalisation calculation* in the saturated GPE (wave turbulence / phonon Boltzmann) showing relaxation to a Bose–Einstein occupation would lift the blackbody character from *deferred* to *structural* and is the single most important theoretical-support item.

2. *A derivation of the absolute saturation pressure* $P_0(\rho_0)$ (the shared Paper II/IV/VII-b closure item), together with the Paper VII-b emergent-expansion factor, would turn T_0 into a falsifiable SSV number — the decisive observational-support item.

3. *A defect-correlation → distortion-spectrum computation* giving numerical μ, y would make §5 testable against COBE/FIRAS [5] and is the cleanest near-term falsification route.

4. *Extending the Kibble–Zurek dynamic range* of `kibble_zurek_gpe` (the Paper VIII Stage 5 simulation) to tighten the exponent, followed by a phonon-production calculation, would test whether any predicted η anchor exists.

7.4 What this paper is not

This paper does not derive: the CMB temperature; the blackbody form of the spectrum; the spectral-distortion amplitudes; the angular power spectrum or its acoustic peaks; a recombination or last-scattering treatment; a primordial gravitational-wave background; or any inflationary, structure-formation, or dark-energy phenomenology. Each belongs to another paper or to a future computation. The paper is a scoped interpretive reading of the CMB as the relaxed primordial phonon bath, plus an explicit ledger of what would be required to make that reading quantitative. Treating it as a complete CMB cosmology would re-introduce exactly the catch-all manifesto drift that the split of the late-series papers was designed to avoid.

8 Conclusion

Within the scoped SSV programme, this paper carries one claim: the cosmic microwave background is the present, red-shifted image of the primordial phonon bath shed by the void-to-saturation transition of Paper VIII — integrated reconnection noise, not a separately postulated radiation relic. The reading is interpretive; Paper VIII does not yet supply a quantitative defect/phonon anchor. The central physical quantities — the temperature, the blackbody form, and the distortions — are explicitly deferred to the named closure inputs rather than claimed.

References

[1] Norland, S., "Saturated Superfluid Vacuum VIII: Cosmogony from the Permissive Void," SSV Series Paper VIII (2026).

[2] Norland, S., "Saturated Superfluid Vacuum (Goldstone supplement): Electromagnetic Propagation Speed and the Photon as a Goldstone Mode," SSV Series (2026).

[3] Norland, S., "Saturated Superfluid Vacuum III: Thermodynamics and Irreversible Time," SSV Series Paper III (2026).

[4] Norland, S., "Saturated Superfluid Vacuum II: Forces as Hydrodynamic Pressure Gradients in the Saturated Superfluid Plenum," SSV Series Paper II (2026).

[5] Fixsen, D. J., *et al.*, "The Cosmic Microwave Background Spectrum from the Full COBE/FIRAS Data Set," *Astrophys. J.* **473**, 576–587 (1996).

A Code and Issue References

Each reference below is pinned so it can be checked directly against the repository (https://github.com/StigNorland/SVT): issues link to their tracker entry, and each script or report links to the exact version — the commit that last modified it. Issue numbers in the text hyperlink to this list.

Issues.

- #10 — https://github.com/StigNorland/SVT/issues/10
- #100 — https://github.com/StigNorland/SVT/issues/100
- #119 — https://github.com/StigNorland/SVT/issues/119
- #129 — https://github.com/StigNorland/SVT/issues/129

Code (each pinned to the commit that last modified it).

- `instruments/paper_viii/kibble_zurek_gpe.py` @ 664a338 — https://github.com/StigNorland/SVT/blob/664a338/instruments/paper_viii/kibble_zurek_gpe.py

Reports (result notes, pinned to their last commit).

- `docs/numerical-minimisation-roadmap.md` @ 77eff54 — https://github.com/StigNorland/SVT/blob/77eff54/docs/numerical-minimisation-roadmap.md

Change record. This paper states its *current* status only. Corrections, withdrawals and reframings are recorded in https://github.com/StigNorland/SVT/blob/main/papers/SSV-IX/CHANGELOG.md, and the full history is in the repository's commits.